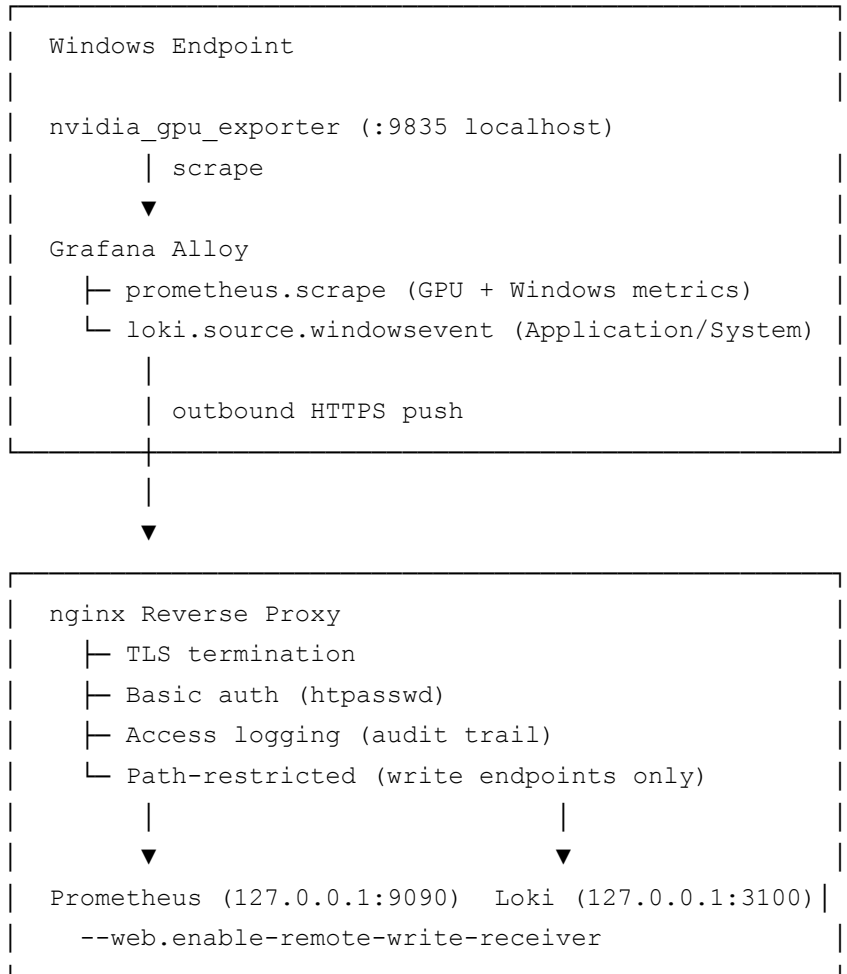


Deploying Grafana Alloy + NVIDIA GPU Exporter on Windows via BigFix

Architecture



Prometheus stays bound to loopback with zero external attack surface. All Windows endpoints push outbound over HTTPS — no inbound scrape ports needed. nginx gives you TLS, auth, rate limiting, and a full audit trail in access logs.

Files

File	Description
<code>Deploy-AlloyAndNvidiaExporter.ps1</code>	PowerShell script — full install of Alloy + GPU exporter
<code>Deploy-AlloyGpuExporter.bes</code>	BigFix Task XML — import into your console
<code>nginx-telemetry-proxies.conf</code>	nginx vhost configs for Prometheus write proxy + Loki push proxy

Components

Component	Version	Service Name	Port	Notes
Grafana Alloy	1.14.1	<code>Alloy</code>	12345 (UI, local)	Telemetry collector
<code>nvidia_gpu_exporter</code>	1.4.1	<code>nvidia_gpu_exporter</code>	9835 (local)	GPU metrics via <code>nvidia-smi</code>
WinSW	2.12.0	—	—	Service wrapper for GPU exporter (MIT license)

Prerequisites

Windows endpoints:

- Windows 10 / Server 2016+ (amd64)
- NVIDIA drivers installed (`nvidia-smi.exe` in PATH or standard location)
- Outbound HTTPS to your nginx proxy (or GitHub for initial binary downloads)

Server side (deploy first):

- nginx with the telemetry proxy vhosts (see `nginx-telemetry-proxies.conf`)
- Prometheus with `--web.enable-remote-write-receiver` flag
- Loki accepting pushes on `/loki/api/v1/push`

- htpasswd files generated for both proxy vhosts
-

Server-Side Setup (Do This First)

1. Enable Prometheus remote write receiver

Add the flag to your Prometheus startup:

```
--web.enable-remote-write-receiver
```

Prometheus stays on `127.0.0.1:9090` — no change to its listen address.

2. Generate htpasswd files

```
# Prometheus write proxy
sudo htpasswd -cB /etc/nginx/.htpasswd-prometheus alloy

# Loki push proxy
sudo htpasswd -cB /etc/nginx/.htpasswd-loki alloy
```

3. Deploy nginx vhosts

Place `nginx-telemetry-proxies.conf` in `/etc/nginx/conf.d/` or integrate into your existing Ansible `nginx_config` role. The file includes commented-out Jinja2 `{% include %}` directives that map to your existing snippet templates for SSL, proxy headers, and security headers. Uncomment those and remove the inline equivalents to match your role structure.

Update `server_name` and SSL cert paths to match your environment, then:

```
sudo nginx -t && sudo systemctl reload nginx
```

4. Test the proxy endpoints

```
# Should return 401 (auth required)
curl -s -o /dev/null -w "%{http_code}" https://prometheus-write.yourdomain.com/ap

# Should return 403 (root path denied)
curl -s -o /dev/null -w "%{http_code}" https://prometheus-write.yourdomain.com/
```

```
# Health check (no auth)
curl https://prometheus-write.yourdomain.com/nginx-health
```

Windows Deployment

Configuration

Open `Deploy-AlloyAndNvidiaExporter.ps1` and update:

```
# Telemetry endpoints (your nginx proxy hostnames)
$PrometheusEndpoint = "https://prometheus-write.yourdomain.com/api/v1/write"
$LokiEndpoint       = "https://loki.yourdomain.com/loki/api/v1/push"

# Basic auth credentials (must match your htpasswd files)
$PrometheusAuthUser = "alloy"
$PrometheusAuthPass = "CHANGE_ME_PROMETHEUS"
$LokiAuthUser       = "alloy"
$LokiAuthPass       = "CHANGE_ME_LOKI"
```

TLS Certificates

If you're using a private CA or self-signed certs on nginx, you have two options:

1. **Distribute the CA cert** to endpoints and set `ca_file` in the Alloy config's `tls_config` blocks (the template has commented placeholders for this).
2. **Use a publicly trusted cert** (Let's Encrypt, etc.) — no CA distribution needed.

Pre-staging Binaries (Air-gapped Networks)

Download and host internally:

- `alloy-installer-windows-amd64.exe` — [Alloy releases](#)
- `nvidia_gpu_exporter_1.4.1_windows_x86_64.zip` — [GPU exporter releases](#)
- `WinSW-x64.exe` — [WinSW v2.12.0](#)

Update the `$AlloyInstallerUrl`, `$GpuExporterUrl`, and `$WinSWUrl` variables.

BigFix Deployment

1. Import `Deploy-AlloyGpuExporter.bes` into your BigFix console.

2. Update the prefetch/ActionScript to reference your hosted PS1.
3. Target using the built-in relevance or your own Computer Group.
4. Deploy — retries 3x at 15-minute intervals on failure.

What Gets Installed on Each Endpoint

Grafana Alloy

Item	Path
Install dir	C:\Program Files\GrafanaLabs\Alloy\
Config	C:\Program Files\GrafanaLabs\Alloy\config.alloy
Data	C:\ProgramData\GrafanaLabs\Alloy\data\
Service	Alloy (auto-start, LocalSystem)
Logs	Event Viewer → Application → "Grafana Alloy"
Registry	HKLM\Software\GrafanaLabs\Alloy

nvidia_gpu_exporter (via WinSW)

Item	Path
Install dir	C:\Program Files\nvidia_gpu_exporter\
Service wrapper	nvidia_gpu_exporter.exe (WinSW renamed)
Service config	nvidia_gpu_exporter.xml
Service	nvidia_gpu_exporter (auto-start)
Logs	C:\Program Files\nvidia_gpu_exporter\logs\
Log rotation	5 MB × 3 files
Failure restart	10s → 30s → stop; resets after 1 hour

Firewall Rules

Rule	Port	Profile	Notes
Grafana Alloy UI	TCP/12345	Domain, Private	Local diagnostics only
NVIDIA GPU Exporter	TCP/9835	Domain, Private	Scraped locally by Alloy

Verification

On the Windows endpoint:

```
Get-Service Alloy, nvidia_gpu_exporter | Format-Table Name, Status

# GPU metrics
Invoke-WebRequest -Uri http://localhost:9835/metrics -UseBasicParsing | Select-Ob

# Alloy UI
Start-Process "http://localhost:12345"

# WinSW status
& "C:\Program Files\nvidia_gpu_exporter\nvidia_gpu_exporter.exe" status
```

On the server (check writes are arriving):

```
# Prometheus - query for a GPU metric
curl -s 'http://127.0.0.1:9090/api/v1/query?query=nvidia_gpu_utilization' | jq .

# Loki - query for Windows event logs
curl -s -G 'http://127.0.0.1:3100/loki/api/v1/query_range' \
  --data-urlencode 'query={job="windows_eventlog"}' \
  --data-urlencode 'limit=5' | jq .
```

Grafana dashboard: Import dashboard ID **14574** for the

utkuozdemir/nvidia_gpu_exporter metric names.

Upgrading

Re-run the BigFix Task with updated version variables in the PS1. The script handles in-place upgrades (stop → uninstall → reinstall → start).

Uninstall

```
# GPU exporter (via WinSW)
$svc = "C:\Program Files\nvidia_gpu_exporter\nvidia_gpu_exporter.exe"
& $svc stop
& $svc uninstall

# Alloy
& "C:\Program Files\GrafanaLabs\Alloy\uninstall.exe" /S

# Cleanup
Remove-Item "C:\Program Files\nvidia_gpu_exporter" -Recurse -Force
Remove-Item "C:\ProgramData\GrafanaLabs" -Recurse -Force
Remove-NetFirewallRule -DisplayName "Grafana Alloy UI"
Remove-NetFirewallRule -DisplayName "NVIDIA GPU Exporter"
```

Security Notes

- **No inbound scrape ports** — all telemetry is pushed outbound from endpoints.
- **Prometheus stays on loopback** — nginx is the only external-facing component.
- **Basic auth** at the proxy layer prevents unauthorized writes.
- **Path-restricted** — nginx only proxies the specific write/push endpoints; root returns 403.
- **TLS everywhere** — Alloy → nginx is encrypted; nginx → Prometheus/Loki is loopback.
- **WinSW over NSSM** — NSSM hasn't been updated since 2017; WinSW is MIT-licensed and actively maintained (commits through March 2026).
- **Audit trail** — nginx access logs capture every write request with source IP and auth user.